

STEPGEN · REQUIREMENTS

StepGen requirements

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DOCUMENT CONTROL

Document ID	SGN-REQ-001	Version	0.5
Status	Draft	Date	2026-09-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/stepgen		
Source	docs/03-requirements.md @ 8182ad3		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.4	2026-09-25	Amish Chadha	TRL 3. Record Amish's decisions (SGN-DDR-001); R11 to SwapCell interface v0.3; new R13 (latch class V1); R12 budget \$650; status of every requirement from SGN-CAL-001
0.3	2026-09-25	Amish Chadha	Concept changed at Amish's direction from a stationary stepper generator to a walking-treadmill vehicle
0.2	2026-09-24	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-24	Amish Chadha	Initial scaffold

StepGen requirements

These are the requirements for the walking-treadmill vehicle at TRL 3. Amish decided the targets for form factor, wheels, speed class, range and budget on 2026-09-25 (SGN-DDR-001), and accepted the TRL 3 recommendations on cost route, belt length, deck rails and key switch the same day (SGN-DDR-002); the rest remain engineering targets to be revised after co-design with users. Every requirement has been checked by calculation in SGN-CAL-001, and Table 1 gives its status. The stepper generator requirements of version 0.2 are withdrawn with that concept.

Table 1. Requirements and status (SGN-CAL-001).

ID	Requirement	Target	Verification	Status at TRL 3
R1	Natural walking on the belt	Usable belt length 1.0 m or more between roller tangents and belt width 0.36 m or more; horizontal force to walk the belt 30 N or less at 5 km/h for an 80 kg rider	Deck geometry in the model; belt drag calculation	At risk: 1.00 m and 22 to 30 N meet the target, but 1.00 m fits a 1.75 m rider only up to 5 km/h; belt kept until co-design (SGN-DDR-002)
R2	Motor runs only while the rider walks	Assist starts only when belt speed stays above 1.5 km/h for 0.5 s; assist power rises with belt speed; motor off within 0.5 s of the belt stopping, either brake lever being pulled or the lanyard coming out	Control logic review; timing budget	Met on paper: 261 ms from belt stop, about 30 ms from a brake lever or the lanyard
R3	Speed limit	Assist cut off at 25 km/h or less, set in firmware; a beginner mode capped at 15 km/h; acceleration limited to 1.0 m/s ² or less (decided by Amish, 2026-09-25)	Controller configuration; dynamics calculation	Met on paper: 32 N m torque limit
R4	Motor power in the pedelec class	Rated continuous motor power 250 W or less (decided by Amish, 2026-09-25)	Motor datasheet	Met on paper
R5	Range on one shared pack	30 km or more at a steady 20 km/h on flat pavement with an 80 kg rider, using the usable energy of one SwapCell pack (decided by Amish, 2026-09-25)	Road-load and energy calculation	Met: 45 km
R6	Hill capability	Climb an 8 % grade at 8 km/h or more with an 80 kg rider, within the motor's rating	Road-load calculation	At risk: 235 W against 250 W
R7	Braking	Two independent hand brakes; stop from 25 km/h within 10 m of brake application on dry pavement, at a deceleration a standing rider can hold (about 3 m/s ²); the belt cannot run forward under the rider's feet when braking	Brake calculation; clutch specification	Met on paper: 8.0 m; 9.6 m on the rear brake alone
R8	Rider can step off and put a foot down	Belt top 250 mm or less above the ground; open sides with no rail higher than the deck along the belt	Massing model	Met: 240 mm
R9	Guard pinch and entanglement points	Belt in-running nips at both rollers, wheel spokes and the rear tire all covered or out of reach of feet, toes and fingers; heel guard between the belt's rear end and the rear tire	Design review against ISO 13857 reach distances	Not verifiable at TRL 3
R10	Size and mass	Length 2.4 m or less, width 0.65 m or less at the handlebar, mass 40 kg or less with the pack	Model; mass estimate	At risk: 2.35 m and 0.61 m met; 37.9 kg leaves 2.1 kg
R11	Shared SwapCell pack	Uses the SwapCell interface v0.3 unchanged: 10 k Ω coding resistor in the INTERLOCK loop (wake, item W), CAN host heartbeat as a vehicle requesting discharge, receiver cradle to the v0.3 envelope; continuous current 15 A or less; every circuit below 60 V DC	Interface review with the SwapCell project	Met on paper: 8.5 A maximum

ID	Requirement	Target	Verification	Status at TRL 3
R12	Low cost and buildable	Parts cost \$650 or less excluding the SwapCell pack on the reference build, which uses a salvaged walking-pad treadmill (items 2 and 3) and a donor 20 in bike (items 7 and 9); new parts are the fallback and are reported but not held to this figure (decided by Amish, 2026-09-25, SGN-DDR-001 and SGN-DDR-002; the pack is priced once in the SwapCell BOM); built with hand tools, a drill press and basic welding	Priced BOM	Met on paper: \$546 (all new \$721)
R13	Pack retention in the vehicle	Receiver cradle meets SwapCell latch class V1 (interface v0.3 item V): no latch release and no power-contact break of 1 ms or longer under 8 g sine and 25 g shocks; pack latch proof 1.72 kN; receiver preload 330 N or more through an over-centre lever with a detent	Paper sizing now; test at TRL 4 (on hold)	Not verifiable at TRL 3

Safety: R2, R3, R7, R8, R9 and R13 are safety requirements for a person standing on a moving belt on a vehicle with a 468 Wh lithium-ion pack. Meeting them on paper does not make the vehicle safe to build or ride; they need test evidence at TRL 4, which is on hold.

■ Assumptions

- Rider 80 kg and 1.75 m; vehicle 35.0 kg without the pack; SwapCell pack 2.85 kg; total 117.9 kg.
- Rolling resistance coefficient 0.010 for 20 x 1.75 in tires on pavement; standing rider drag area CdA about 0.70 m²; air density 1.2 kg/m³.
- Motor and gearbox efficiency 80 %, controller efficiency 95 %, 3 W of auxiliary load.
- Usable pack energy 411 Wh: the SwapCell sizing note gives 457 Wh at the terminals per full cycle, and a 10 % reserve is kept.
- Walking speed on the belt 4 to 6 km/h; belt rolling-bed coefficient about 0.02.
- The full list is in SGN-CAL-001, Table 1.

■ Requirements not met or at risk

- **R1 is at risk.** The 1.0 m target is met exactly, but a 1.75 m rider at 6 km/h, or a 1.90 m rider at 5 km/h, needs more belt. A longer belt breaks the 2.4 m length of R10. Decided by Amish, 2026-09-25 (SGN-DDR-002): keep the 1.05 m roller pitch and revisit after co-design.
- **R6 is at risk.** The 8 % climb needs 235 W at the wheel, a 6 % margin to the 250 W rating.
- **R10 is at risk on mass.** 37.9 kg with the pack leaves 2.1 kg, less than a 10 % growth allowance. The 60 x 30 x 2 mm deck rails (SGN-DDR-002) added 1.08 kg to close the weld fatigue concern.
- **R12 depends on salvage.** It is met on paper at \$546 on the salvage reference build; the \$40 and \$50 salvage prices are estimates, and an all-new build (\$721) would be \$71 over.
- **R9 and R13** can only be verified on hardware, which is TRL 4 work and on hold by Amish's instruction.
- **Legal classification is unresolved.** R3 and R4 follow EU pedelec limits, but a vehicle without pedals may not qualify as a pedelec in the EU or as an e-bike in the US (see SGN-PRB-001). Amish decided on 2026-09-25 that the category is confirmed in a first target country before any road use; the country is still open.